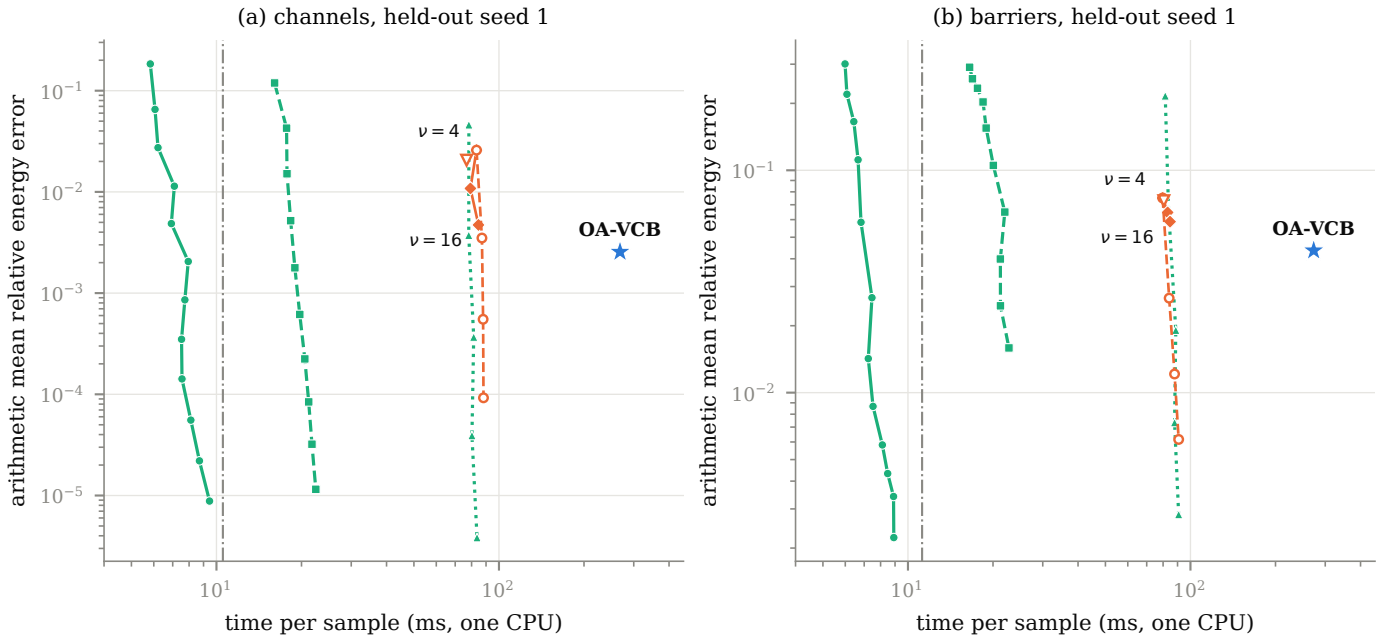
(a) channels, held-out seed 1
(b) barriers, held-out seed 1
arithmetic mean relative energy error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
$\nu = 4$
$\nu = 16$
OA-VCB
10^{1}
10^{2}
time per sample (ms, one CPU)
OA-VCB, one pass
classical ref., one pass, Chebyshev $\nu = 4, 8, 16$
classical ref., one pass, symm. Gauss–Seidel
classical ref. ($\nu = 4$) applied 1–4 times
two-grid PCG, 1, 2, ... iterations
SA-AMG PCG, 1, 2, ... iterations
PCG with classical V(4,4), 1, 2, ...
sparse direct solve (time only; error below range)